Answers and talk notes

Determine Surface Area of Pyramids

What your student is learning

Tonight we find LATERAL area — the slanted faces of a pyramid only, leaving the base out on purpose.

Answers

Ask your student to explain before you look.

1. **B.** The area of the slanted triangle faces only. The area of the slanted triangle faces only is the lateral area.
2. **D.** 56 square units. 56 square units: 4 × 14 = 56.
3. **A.** 40 square units. 40 square units: one side is ½ × 4 × 5 = 10, and 4 × 10 = 40.
4. **C.** 56. 56: lateral area plus base area is 40 + 16 = 56.
5. **A.** Lateral area. Lateral area is the area of the sides only, which is what you paint.

Talk about it

Why do we leave the base out when we only paint or cover the slanted sides?

Home mission

Find a gift box and a lampshade or paper cup. For each one, ask: is the bottom covered? Say when you would add the base and when you would leave it out.

When they are stuck, ask

- Is the bottom being covered in this problem?
- How many slanted faces does this pyramid have?
- Are you using the slant height, or the height through the middle?

Try not to say

- Surface area always includes the base.
- Lateral area and surface area are the same thing.
- Any height measurement works in the triangle formula.